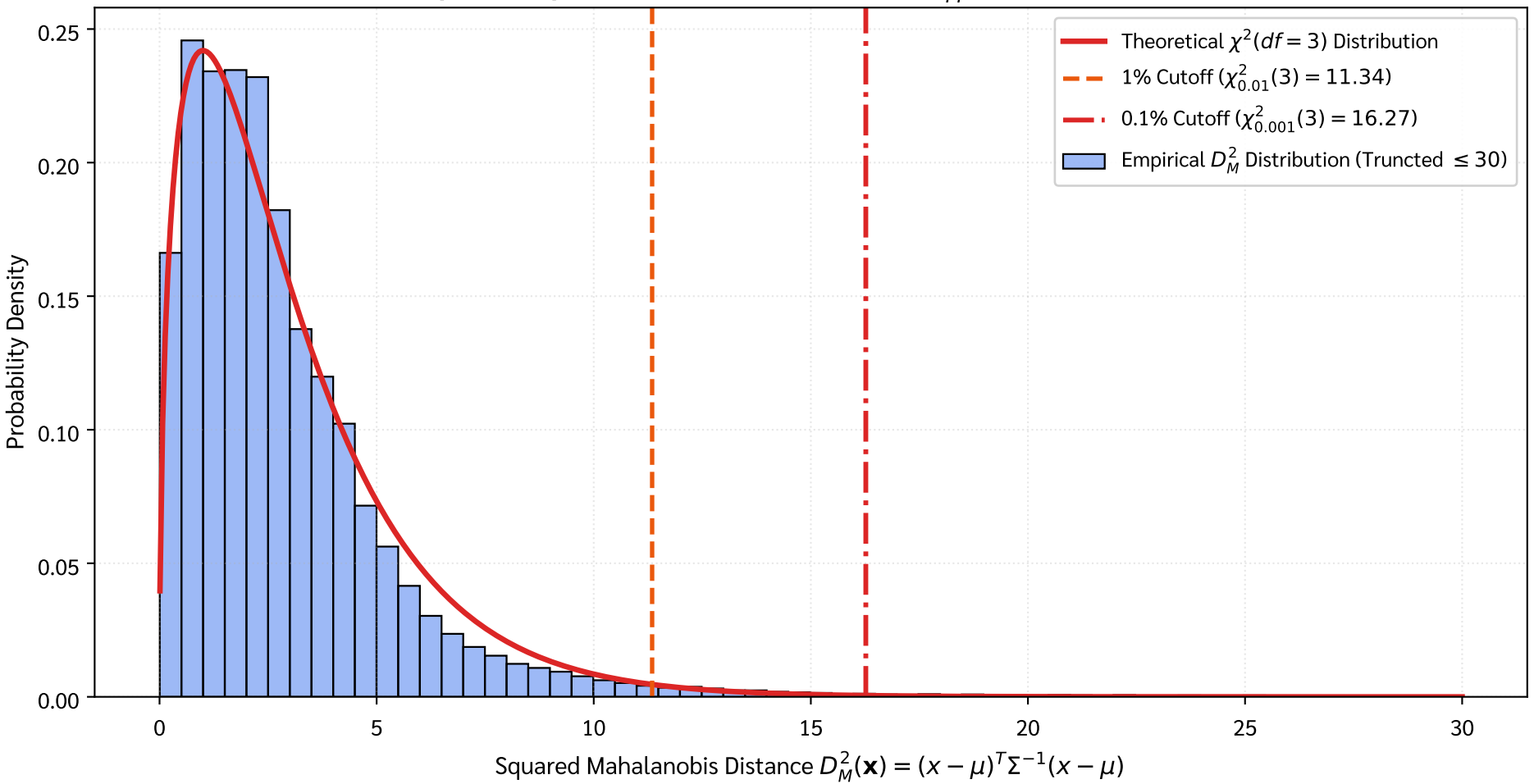
Figure 2: Empirical Squared Mahalanobis Distance D_M^2 vs. Theoretical $\chi^2(3)$ Density
Theoretical $\chi^2(df = 3)$ Distribution
1% Cutoff ($\chi^2_{0.01}(3) = 11.34$)
0.1% Cutoff ($\chi^2_{0.001}(3) = 16.27$)
Empirical D_M^2 Distribution (Truncted ≤ 30)
Probability Density
0.25
0.20
0.15
0.10
0.05
0.00
0
5
10
15
20
25
30
Squared Mahalanobis Distance $D_M^2(\mathbf{x}) = (x - \mu)^T \Sigma^{-1} (x - \mu)$